

# HODARI-SADIKI HUBBARD-JAMES

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Department of Physics and Astronomy, Agnes Scott College

141 E College Ave ◊ Decatur, GA 30030

## RESEARCH INTERESTS

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**Stellar Activity and Ages, Astrobiology, and Time-Domain Astronomy.** My research explores three interconnected areas that advance our understanding of life in the universe. I characterize nearby K dwarf stars through spectroscopy to identify older, quieter systems that could host habitable planets. I investigate astrobiology with focus on photosynthesis under non-solar light sources, particularly how life might adapt to K and M dwarf stars. I also conduct optical and radio follow-up of unexplained astrophysical transients to constrain their origin and physical mechanisms.

## ACADEMIC APPOINTMENTS

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**Assistant Professor of Astronomy**

*Agnes Scott College*

*July 2023- Present*

*Decatur, GA*

**Co-Director**

*Bradley Observatory at Agnes Scott College*

*July 2024- July 2026*

*Decatur, GA*

## EDUCATION

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**Ph.D. in Astronomy**

*Georgia State University*

*2023*

*Atlanta, GA*

**Dissertation:** Spectral Characterization of a Complete Equatorial Sample of 615 K Dwarfs

**Advisor:** Todd J. Henry

**M.S. in Physics with Astronomy Concentration**

*Georgia State University*

*2020*

*Atlanta, GA*

**M.S. in Biomedical Engineering**

*University of Cincinnati*

*2014*

*Cincinnati, OH*

**B.A. in Physics minor Biology**

*Berea College*

*2012*

*Berea, KY*

## GRANTS & EXTERNAL FUNDING

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**Funded Awards (\$448,669 awarded)**

**NASA COMPASS: Stellar Activity in Nearby K Dwarfs (\$375,256)**

*2026–2028*

National Aeronautics and Space Administration *Principal Investigator.*

**NASA Space Grant – AsTropaLooza (\$27,000)**

*Spring 2025*

Georgia Space Grant Consortium (NASA) *Co-Investigator.*

**SREB & Provost's Dissertation Year Awards (\$35,000)**

*2022–2023*

Southern Regional Education Board & Georgia State University

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|----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| <b>Holder Professional Development Award (\$5,000)</b><br>ISLANDS Astrobiology Lab at the Bradley Observatory                                      | 2026–2027                                                                                              |
| <b>Professional Development Committee (PDC) Award (\$2,600)</b><br>Astrobiology at Agnes: Research and Conference Engagement                       | 2025–2026                                                                                              |
| <b>Professional Development Committee (PDC) Award (\$3,813)</b><br>IAU General Assembly, Cape Town, South Africa                                   | 2024–2025                                                                                              |
| <b>Pending &amp; In Process</b>                                                                                                                    |                                                                                                        |
| <b>NASA Community Project Funding – Bradley Observatory (\$683K–\$1.4M)</b><br>U.S. Senate, Offices of Sen. Raphael Warnock<br>and Sen. Jon Ossoff | FY2027<br><i>Lead author. Advanced to Senate Appropriations<br/>by both Georgia senators; pending.</i> |

## AWARDS & HONORS

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|----------------------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>NAS Frontiers of Science Fellow</b><br>National Academy of Sciences                                                                 | 2026      |
| <b>IAU-OAD Recommended List: <i>Stars Without Borders</i></b><br>International Astronomical Union, Office of Astronomy for Development | 2026      |
| <b>Beth Brown Memorial Award (Honorable Mention)</b><br>National Society of Black Physicists (NSBP)                                    | Nov. 2022 |
| <b>Chambliss Astronomy Achievement Award at AAS 240</b><br>American Astronomical Society                                               | Jun. 2022 |
| <b>Most Inspirational Alumnus Award</b><br>Berea College African Students Association                                                  | 2016      |
| <b>Waldemar Noll Prize in Physics</b><br>Berea College                                                                                 | 2012      |

## PUBLICATIONS \* \*

 **ORCID:0000-0003-4568-2079**

### First-Author & Lead Publications

1. The Solar Neighborhood LV: Spectral Characterization of an Equatorial Sample of 580 K Dwarfs  
**Hubbard-James, H.-S., Carrazco Gaxiola, S., ... Arbogast, A. (Student Mentee), et.al.,**  
2026, AJ 171 175.
2. Spectral Characterization of a Complete Equatorial Sample of 615 K Dwarfs  
**Hubbard-James, H.-S.,** 2024, Proceedings of the International Astronomical Union, 20(S393),  
144-149.
3. The Solar Neighborhood L: Spectroscopic Discovery of K Dwarfs Younger than 1 Gyr and New  
Binaries within 30 Parsecs  
**Hubbard-James, H.-S.,** Lesley, D.X., Henry, T.J., et al., 2022, AJ 164 174.

## Contributing Author Publications

4. Visual Orbits of Spectroscopic Binaries with the CHARA Array. V. HD 210763 and HD 221950  
Lester K. V., Singleton J., ... **Hubbard-James, H.-S.**, et al., 2026, AJ, in press.
5. Arecibo Wow! II: Revised Properties of the Wow! Signal from Archival Ohio SETI Data  
Méndez A., Ortiz Ceballos K. N., Zuluaga J. I., ... **Hubbard-James, H.-S.**, **Le, M. (Student Mentee)**, et al., 2026, ApJ, Submitted.
6. A Radio Loud Quiescent K Dwarf  
Frail, D., Hyman, S., Silverstein, M., ... **Hubbard-James, H.-S.**, **Byam, J. (Student Mentee)**, et al., 2025, ApJ, 989, 186.
7. Mind the Gap. I. H $\alpha$  Activity of M Dwarfs Near the Partially/Fully Convective Boundary and a New H $\alpha$  Emission Deficiency Zone on the Main Sequence  
Jao, W.-C., Henry, T. J., White, R. J., ... **Hubbard-James, H.-S.**, et al., 2023, AJ, 166, 63.
8. Visual Orbits of Spectroscopic Binaries with the CHARA Array. IV. HD 61859, HD 89822, HD 109510, and HD 191692  
Lester K. V., Schaefer G. H., ... **Hubbard-James, H.-S.**, et al., 2022, AJ, 164, 228.
9. The Solar Neighborhood XLVIII: Nine Giant Planets Orbiting Nearby K Dwarfs, and the CHIRON Spectrograph's Radial Velocity Performance  
Paredes L. A., Henry T. J., Quinn S. N., Gies D. R., Hinojosa-Góñi R., **James H.-S.**, Jao W.-C., et al., 2021, AJ, 162, 176.
10. Mapping out the Stellar Populations of IC 2602 and IC 2391  
Nisak, A.H., White, R.J., Yep, A., ... **James, H.-S.**, et al., 2022, AJ, 163, 278.
11. TOI 564 b and TOI 905 b: Grazing and Fully Transiting Hot Jupiters Discovered by TESS  
Davis A. B., Wang S., Jones M., Eastman J. D., ... **James, H.-S.**, et al., 2020, AJ, 160, 229.
12. TOI 694b and TIC 220568520b: Two Low-mass Companions near the Hydrogen-burning Mass Limit Orbiting Sun-like Stars  
Mireles I., Shporer A., Grieves N., Zhou G. ... **James, H.-S.**, et al., 2020, AJ, 160, 133.
13. KELT-25 b and KELT-26 b: A Hot Jupiter and a Substellar Companion Transiting Young A Stars Observed by TESS  
Rodríguez Martínez R., Gaudi B. S., ... **James, H.-S.**, et al., 2020, AJ, 160, 111.
14. A Well-aligned Orbit for the 45 Myr-old Transiting Neptune DS Tuc Ab  
Zhou G., Winn J. N., Newton E. R., Quinn S. N., ... **James, H.-S.**, et al., 2020, ApJL, 892, L21.
15. TESS Spots a Hot Jupiter with an Inner Transiting Neptune  
Huang C. X., Quinn S. N., Vanderburg A., ... **James, H.-S.**, et al., 2020, ApJL, 892, L7.
16. MASCARA-4 b/bRing-1 b: A retrograde hot Jupiter around a bright A-type star  
Dorval P., Talens G. J. J., Otten G. P. P. L., ... **James, H.-S.**, et al., 2020, A&A, 635, A60.
17. TESS discovery of an ultra-short-period planet around the nearby M dwarf LHS 3844  
Vanderspek R., Huang C. X., Vanderburg A., Ricker G. R., Latham D. W., Seager S., Winn J. N., ... **James, H.-S.**, et al., 2019, ApJL, 871, L24.
18. HD2685 b: A Hot-Jupiter orbiting an early F-type star detected by TESS  
Jones M. I., Brahm R., Espinoza N., Wang S., ... **James, H.-S.**, et al., 2019, A&A, 625, A16.
19. HD 202772A b: A Transiting Hot Jupiter around a Bright, Mildly Evolved Star in a Visual Binary Discovered by TESS  
Wang S., Jones M., Shporer A., Fulton B. J., ... **James, H.-S.**, et al., 2019, AJ, 157, 51.

## TEACHING EXPERIENCE

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### **Instructor of Record**

Agnes Scott College

*Fall 2023- Present*

Decatur, GA

### **Astronomy Courses:**

- AST 120: Introduction to Astronomy (The Solar System) (*Fall 2026*)
- AST 200L: Intermediate Observational Techniques (*Spring 2024, 2025, 2026*)
- AST 300: Astrophysics I: Radiation (*Fall 2023*)
- AST 301: Astrophysics II: Dynamics (*Fall 2024, 2026*)
- GBL 102/103: Global Journeys: Astronomy and Local Communities– Puerto Rico (*Spring 2026, 2027*)

### **Physics & Mathematics Courses:**

- PHY/MAT 131: Intro to Computer Programming (*Fall 2023 & Spring 2025*)
- PHY/MAT 231: Think Like a Data Scientist (*Spring 2024, 2026, 2027*)
- PHY 205: Math for Physicists & Engineers (*Spring 2025*)
- PHY 400: Physics Capstone Colloquium (*Fall 2024, 2026*)
- PHY 420: Advanced Seminar in Physics & Astronomy (*Fall 2024, 2026*)
- PHY 503L: Elements of Physics II Lab (*Spring 2024, 2025, 2026*)

### **Instructor of Record**

Freedom University

*2018- 2019, Fall 2025*

Atlanta, GA

- STEM: Astronomy (*Fall 2025*) – virtual introductory astronomy course
- SAT Mathematics Preparation (*2018 & 2019*)

### **Joint Instructor**

Georgia State University

*Spring 2022*

Atlanta, GA

- Co-Lecturer for a Synchronous (Online and In-Person Instruction) Introductory Astronomy Course  
ASTR 1000 (Introduction to the Universe)

### **Graduate Teaching Assistant**

Georgia State University

*Aug 2017 - Jul 2021*

Atlanta, GA

- Instructor for Introductory Astronomy Lab ASTR 1010 (Astronomy of the Solar System)  
Fall 2017, 2018, 2019, & 2020, Spring 2019, & 2021
- Instructor for Introductory Astronomy Lab ASTR 1020 (Stellar and Galactic Astronomy)  
Fall 2020, Spring 2018, & 2020

## STUDENT MENTORING & SUPERVISION

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### Current Undergraduate & Post-baccalaureate Mentees:

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| <b>Makaila Jennings</b> - Undergraduate Student, Agnes Scott College.<br><i>Research focus: Astrobiology and photosynthesis under non-solar illumination</i><br><i>Placement: NSBP Simmons Research Program, Summer 2026</i>                                  | Jan 2026 - Present |
| <b>Lia Faulks</b> - Undergraduate Student, Agnes Scott College.<br><i>Research focus: Astrobiology and photosynthesis under non-solar illumination</i>                                                                                                        | Jan 2026 - Present |
| <b>Kayla Gossett Roper</b> - Undergraduate Student, Agnes Scott College.<br><i>Research focus: Astrobiology and photosynthesis under non-solar illumination</i>                                                                                               | May 2025 - Present |
| <b>Sasha Arbogast</b> - Undergraduate Student, Agnes Scott College.<br><i>Research focus: Radial velocity confirmation of K dwarf companions and exoplanets</i><br><i>Co-author on AJ publication; Cornell REU, Roman Coronagraph scheduling, Summer 2026</i> | May 2025 - Present |
| <b>Mai Le</b> - Post-baccalaureate Researcher, Agnes Scott College.<br><i>Research focus: Optical archival analysis of the Wow! Signal field</i><br><i>Co-author on ApJ submission</i>                                                                        | Aug 2024 - Present |

### Current Graduate Student Mentees:

|                                                                                                                                                                                                |                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| <b>Sebastian Carrazco Gaxiola</b> - PhD Candidate, Georgia State University.<br><i>Research focus: Stellar spectroscopy and K dwarf characterization</i><br><i>Co-author on AJ publication</i> | Aug 2022 - Present |
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### Former Mentees:

|                                                                                                           |                      |
|-----------------------------------------------------------------------------------------------------------|----------------------|
| <b>Jacinda Byam</b> - Undergraduate Student, College of Charleston<br><i>Co-author on ApJ publication</i> | May 2024 - Aug 2025  |
| Nia Suitt - Graduate Student, University of Central Florida.                                              | Jan 2024 - May 2025  |
| Daniela Garcia-Lara - Undergraduate Student, Agnes Scott College.                                         | May 2024 - May 2025  |
| Tim Johns - Graduate Student, Georgia State University.                                                   | May 2022 - July 2023 |
| D. Xavier Lesley - Graduate Student, Ohio State University.<br><i>Co-author on AJ publication</i>         | Jan 2021 - July 2023 |
| Dan Johns - Graduate Student, Georgia State University.                                                   | Sep 2020 - May 2022  |
| Andrey Brevett - Undergraduate Student, Berea College.                                                    | Aug 2018 - Aug 2022  |
| Edgar Ortiz - High School Student, Freedom University.                                                    | Aug 2015 - Aug 2016  |
| Chukwuemeka Chikelu - Graduate Student, University of Cincinnati.                                         | Aug 2014 - Aug 2015  |

## SERVICE & PROFESSIONAL ACTIVITIES

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### Professional Service

**Conference Programming Committee, NSBP/NSHP Joint National Meeting** 2026–2027  
National Society of Black Physicists / National Society of Hispanic Physicists *Co-organizing ASTRO sessions for the joint national meeting, Raleigh, NC.*

### SARA Institutional Representative

Southeastern Association for Research in Astronomy (SARA)

2024–Present

**Co-Organizer, AAS Division on Dynamical Astronomy Meeting** 2024–2025  
 Georgia Institute of Technology *Assisted in planning and organizing national scientific meeting held at Georgia Tech.*

**Judge, NSBP/NSHP Joint National Meeting** Nov. 2024  
 National Society of Black Physicists / National Society of Hispanic Physicists *Served as judge for undergraduate and graduate research presentations.*

**Astronomy Curriculum Committee Member** Sept 2021 - June 2023  
 Georgia State University

**CHIRON Data Manager** Dec 2018 - June 2023  
 RECONS/Georgia State University

### **Institutional Service**

**NCAA III - Faculty Athletics Representative (FAR)** Sept. 2024- Present  
 Agnes Scott College

**Faculty Advisor, Agnes Scott College SPS** Aug 2023 - Aug 2025  
 Agnes Scott College

**Research Advisor, STEM Scholars Program** Summer 2024 & 2025  
 Agnes Scott College

**Physics Faculty Search Committee** Aug 2023 - May 2024  
 Agnes Scott College

### **Outreach & Public Engagement**

**Co-Organizer, AsTropaLooza: The Astronomy of ATL Showcase** Spring 2025  
 Bradley Observatory, Agnes Scott College *Public outreach event funded by NASA Space Grant with student-led programming.*

**Organizer & Participant, Planetary Reach Workshop** Fall 2024  
 Bradley Observatory, Agnes Scott College *Workshop focused on developing equitable STEM outreach strategies for Black and Brown youth.*

**STEM Hands Summer Camp** Jun 2022  
 Georgia Outreach Team for Space (GOT Space)

**NASA Science Activation Planetary Reach Workshop** Apr 2022  
 Arizona State University

## **PROFESSIONAL AFFILIATIONS**

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**International Astronomical Union (IAU)** Aug 2023 - Present

**National Society of Black Physicists (NSBP)** Aug 2021 - Present

**American Astronomical Society (AAS)** Oct 2017 - Present

**REsearch Consortium On Nearby Stars (RECONS)** Aug 2017 - Present

**Southern Regional Education Board (SREB)** Aug 2022 - Aug 2023

**Biomedical Engineering Society (BMES)** May 2015 - May 2017

# TECHNICAL PROFICIENCY

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## Programming Languages

- Expert: Python, Matlab, bash
- Intermediate Proficiency: IDL, R, HTML/CSS
- Limited Proficiency: C++, Julia

## Software and Packages

- Astronomy: Astropy, IRAF/PyRAF, TOPCAT, DS9
- Telescope Operation: CTIO 1.5 telescope (computerized), CTIO 0.9 telescope (computerized), Multiple Tripod & Table Top Telescopes (manual)
- Others: LaTeX, MS Office, Git Repository
- AI & Agentic Tools: Claude Code, OpenAI Codex, and LLM APIs; build and deploy agent-based workflows for research pipelines, data analysis, and course material development

# INVITED TALKS & PRESENTATIONS

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|------------------------------------------------------------------------------------------------------|------------|
| <b>Commencement Address</b><br>Freedom University                                                    | May 2026   |
| <b>Invited Talk: Finding the K Dwarf Advantage, RECONSaganza</b><br>Georgia State University         | Aug. 2026  |
| <b>Invited Talk: Dept Physics &amp; Astronomy Colloquium</b><br>Georgia State University             | July. 2025 |
| <b>Invited Talk: Planetary Science &amp; Astrobiology Seminar</b><br>Georgia Institute of Technology | Oct. 2024  |
| <b>Invited Talk: Dept Physics &amp; Astronomy Colloquium</b><br>Georgia College & State University   | June 2024  |
| <b>Invited Talk at the Bradley Observatory Open House</b><br>Agnes Scott College                     | Dec. 2022  |

# SELECTED CONFERENCE PRESENTATIONS

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1. The K Dwarf Advantage: Stellar Activity, Ages, and the Search for Habitable Worlds  
**Hubbard-James, H.-S.**, 2026, Poster presentation, Kavli Frontiers of Science Symposium, National Academy of Sciences, Irvine, CA.
2. Spectral Characterization of a Complete Equatorial Sample of 615 K Dwarfs  
**Hubbard-James, H.**, Gaxiola, S., Henry, T., Lesley, D.X., Paredes, L. A., Nisak, A., 2024, Poster presentation, 32nd General Assembly International Union (IAUGA 2024).
3. Spectral Characterization of a Complete Equatorial Sample of 615 K Dwarfs  
**Hubbard-James, H.**, Gaxiola, S., Henry, T., Lesley, D.X., Paredes, L. A., Nisak, A., 2024, Oral presentation, 2024 Astrobiology Science Conference (AbSciCon 2024).
4. Spectral Characterization of a Complete Equatorial Sample of 665 K Dwarfs within 33 Parsecs — Active Stars, Calm Stars, and the Best Places for Habitable Worlds  
**Hubbard-James, H.**, 2023, Oral presentation, American Astronomical Society 242nd Meeting.

5. Spectroscopic Identification of Young and Active K Dwarfs Within 25 Parsecs  
**Hubbard-James, H.**, Lesley, D.X., Henry, T.J., Paredes, L. A., Nisak, A., 2022, Poster presentation, American Astronomical Society 240th Meeting Abstracts.
6. Spectroscopic Identification of Young and Active K Dwarfs Within 25 Parsecs  
**Hubbard-James, H.**, Lesley, D.X., Henry, T.J., Paredes, L. A., Nisak, A., 2022, Oral presentation, 2022 Astrobiology Science Conference (AbSciCon 2022).
7. Spectroscopic Identification of Five K Dwarfs Younger than 1 GYR Within 30 parsecs  
**Hubbard-James, H.**, Lesley, D.X., Henry, T.J., 2021, Oral presentation, 2021 National Society of Black Physicists Conference.

## OTHER PROFESSIONAL EXPERIENCE

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|                                                                               |           |
|-------------------------------------------------------------------------------|-----------|
| GRE and SAT Instructor, University of Georgia/Educational Testing Consultants | 2019-2020 |
| Lab Technologist, Labsolutions                                                | 2017      |
| Biomedical Engineer, Castle Medical                                           | 2015-2016 |
| Graduate Research Assistant, University of Cincinnati                         | 2012-2014 |

## PROFESSIONAL REFERENCES

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|                                                                   |                              |
|-------------------------------------------------------------------|------------------------------|
| <b>Todd J. Henry, PhD</b>                                         | <i>Doctoral Advisor</i>      |
| Distinguished Professor of Astronomy, Georgia State University    | thenry88@gsu.edu             |
| Director, REsearch Consortium On Nearby Stars (RECONS) and SMARTS |                              |
| <b>Abel Méndez, PhD</b>                                           | <i>Research Collaborator</i> |
| Professor, University of Puerto Rico at Arecibo                   | abel.mendez@upr.edu          |
| Director, Planetary Habitability Laboratory                       |                              |